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Workshop on Artificial Intelligence and Machine Learning Applications

PUCP

The Cost of the Prompt: Prompt-Free SAM 2 on Sentinel-2 Fields

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MOTIVATION & RESEARCH PROBLEM

- SAM 2 [1] segments anything from **one human click**. Mapping thousands of fields needs it **prompt-free**: the automatic generator clicks a regular point grid instead.
- Sentinel-2 parcels are far from SA-1B [2] objects: 10 m pixels, a median parcel of **113 px (≈ 11 × 11 px)**.

Question: how well does frozen, prompt-free SAM 2 find agricultural parcels, and what limits it: the model or the prompt?

OBJECTIVES / CONTRIBUTION

- **Teardown** of frozen SAM 2.1: parameters, compute per block, latency.
- **Baseline** of the automatic generator against parcel ground truth.
- **Adapt** the mask decoder (encoder frozen) and test it with the deployment pipeline.
- **Diagnose** the gap: domain shift (Spain vs. France) and prompt-grid density.

METHODOLOGY

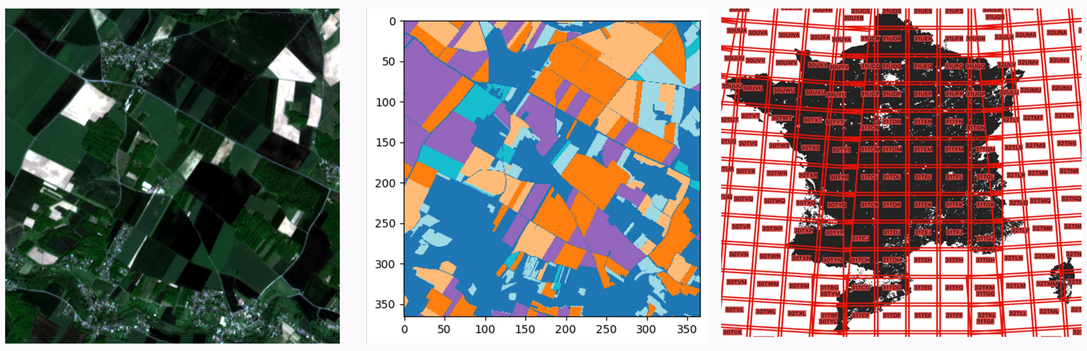


Fig. 1. A test patch (RGB) and its parcel map; the 6 Sentinel-2 tiles lie in Catalonia and France.

- **Data:** Sen4AgriNet [3], 211,375 parcels (≥ 20 px) in 888 Sentinel-2 patches (366 × 366 px, RGB, 10 m, May), 6 tiles, Spain and France. Split 70/15/15 by patch, stratified by tile, seed 42; test = 132 patches.
- **Model:** SAM 2.1 Hiera-Small; automatic generator, 16 × 16 grid (pps = 16), predicted-IoU ≥ 0.5, stability ≥ 0.8.
- **Metric:** for each true parcel, best IoU over all generated masks; mean over parcels.
- **Adaptation (v1):** train decoder transformer layer 2 + IoU head (1.71 M params, 3.7 %), Dice loss, one centroid prompt per parcel.
- **Ablation (v2):** prompt (centroid / random / grid) × output tokens (0 / 1–3), checkpoint chosen by generator IoU; grid sweep pps 16 / 32 / 64.

CONCLUSIONS

- **The bottleneck is prompt density, not the decoder:** a 16 × 16 grid reaches only 32% of parcels; a 64 × 64 grid raises IoU 0.083 → 0.185, while no decoder fine-tune beat the frozen model.
- **Validate with the deployment pipeline:** tuning on centroid prompts improved validation (0.312 → 0.530) and broke the automatic generator (0.156 → 0.064).
- **Removing the human prompt has a price:** every grid point is a decoder pass, so the better grid costs 0.98 → 17.72 s per patch.

Limits: one seed; per-patch aggregates; grid sweep on 40 Spanish test patches; the Spain–France gap mixes parcel size with crop and radiometric shift.

KEY REFERENCES

[1] N. Ravi et al., “SAM 2: Segment Anything in Images and Videos,” arXiv:2408.00714, 2024.
[2] A. Kirillov et al., “Segment Anything,” in Proc. IEEE/CVF ICCV, pp. 4015–4026, 2023.
[3] D. Sykas et al., “A Sentinel-2 Multiyear, Multicountry Benchmark Dataset for Crop Classification and Segmentation With Deep Learning,” IEEE JSTARS, vol. 15, pp. 2937–2953, 2022.

RESULTS & DISCUSSION

A. The prompt dominates compute. The encoder costs 134 GMAC per image, the decoder 1.4 GMAC per prompt: a 16 × 16 grid (256 prompts) already costs **2.6 × the encoder**, 32 × 32 ≈ 10 ×.

Frozen baseline: mean best IoU **0.161**; 15% of parcels with IoU > 0.5; Spain 0.085 ≪ France 0.220. Masks often merge neighbouring parcels.

B. Fine-tuning helped the proxy, hurt the pipeline. Validation with centroid prompts rose 0.312 → 0.530; the automatic generator on test fell 0.156 → **0.064** (paired Δ -0.092, 95 % CI [-0.103, -0.083]; masks per patch 78.7 → 41.1). **Training prompt ≠ inference prompt.**

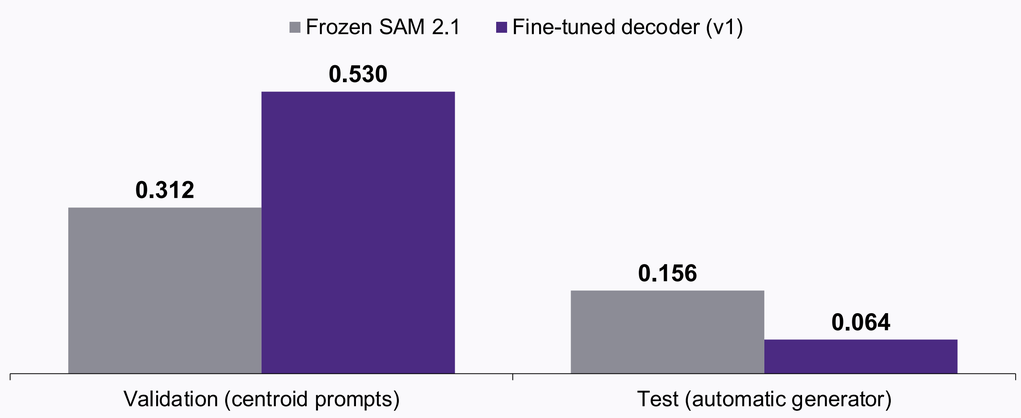


Fig. 2. Mean best IoU before / after decoder fine-tuning (v1), on the proxy and on the real pipeline.

C. The grid misses the fields. At pps = 16 (a point every 22.9 px) only **32%** of parcels get a grid point (Spain 24%, France 40%; 20–49 px parcels 6%); pps = 32: 68%, pps = 64: **95%.** Patch IoU tracks parcel size (Spearman ρ = 0.87; confounded with country). Coverage is an upper bound, not IoU.

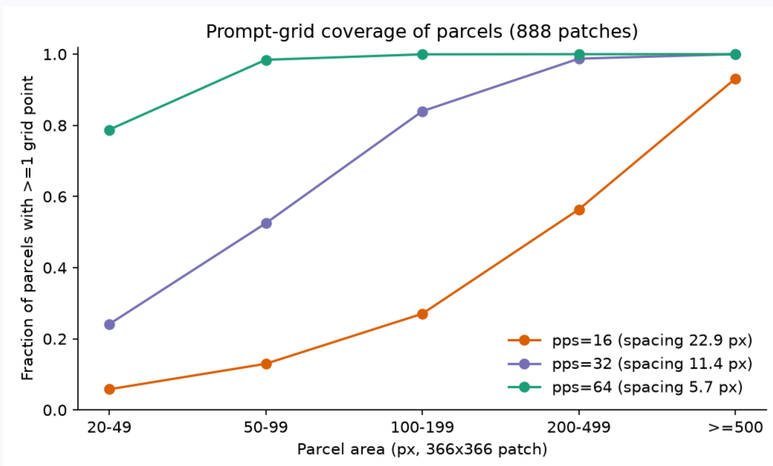


Fig. 3. Fraction of true parcels hit by ≥ 1 grid point, by parcel size (all 888 patches).

D. Denser grid, better masks (v2). On 40 test patches (Spain), IoU rises 0.083 → 0.146 → **0.185** for pps 16 / 32 / 64, at 0.98 → 8.86 → 17.72 s per patch. Filters off (pps = 16): [FILTERS_OFF_IOU].

Decoder ablation: with checkpoints chosen by generator IoU, no trained variant beat the frozen model (best = epoch 0, 0.156); choosing by validation loss gives 0.092.

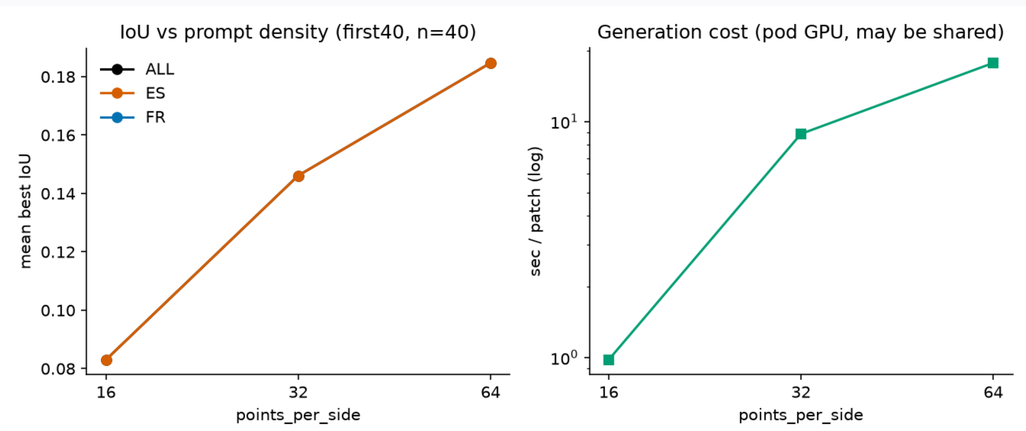


Fig. 4. Mean best IoU (left) and seconds per patch (right, log scale; shared pod GPU) vs. grid density, first 40 test patches.

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CONTINUE THE CONVERSATION

Contact: andre.rubio@pucp.edu.pe
Code, splits and results:
github.com/ZagonTM/Sen4Sam

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